

Whole Exome Sequencing Analysis

Patient name: XX	PIN: XX
Age: 52 Years	Sex: Female
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: NA
Specimen: XX	Sample Received Date: NA
	Report Date: NA

Clinical History

Proband, XX has a diagnosis of atypical retinitis pigmentosa, which subsequently progressed to retinal detachment and resulted in complete blindness. XX daughter, XX [NA, 2025], is currently asymptomatic with no visual concerns and has an ongoing pregnancy at 12 weeks and 5 days of gestation. Proband, XX has been evaluated for pathogenic variations

Results

Likely Pathogenic variant was identified in MFSD8 gene

List of significant variant identified:

Gene	Variant*	Allele Status	Region	Disease	Inheritance	Classification*
#MFSD8 (-)	c.1361T>C (p.Met454Thr)	Homozygous	Exon 12	Ceroid lipofuscinosis, neuronal, 7 (OMIM# 610951) Macular dystrophy with central cone involvement (OMIM# 616170)	Autosomal Recessive	Likely Pathogenic

#This homozygous likely pathogenic variant (c.1361T>C, p.Met454Thr) in MFSD8 gene in XX has been detected in XX (NA, 2025) in heterozygous state.

*Genetic test results are based on the recommendation of American college of Medical Genetics [1]. No other variant that warrants to be reported for the given clinical indication was identified

Variant Interpretation

Gene	Variant*	Inheritance	Disorder or Phenotype	Criteria	Classification*
<i>MFSD8</i> (-)	c.1361T>C (p.Met454Thr)	Autosomal Recessive	Ceroid lipofuscinosis, neuronal, 7 (OMIM# 610951) Macular dystrophy with central cone involvement (OMIM# 616170)	PM2, PM3, PP3, PP5	Likely Pathogenic
Region	Allele State	Allelic Read Depths			
Exon 12	Homozygous	Ref(A): 0, Alt(G): 131, VAF: 100.00%			
Genomic Position			Variant Frequency		
Chr4:NC_000004.12:g.127920826A>G (NM_001371596.2)			0.071% max frequency observed in Annotated gnomAD v4 SouthAsian		

Variant summary: A homozygous missense variation in exon 12 of *MFSD8* gene (Chr4:NC_000004.12:g.127920826A>G; NM_001371596.2; Depth: 128x) that results in amino acid substitution of Threonine for Methionine at codon 454 (p.Met454Thr) was detected.

Population Frequency: This variant has been observed in 65/91048 (0.071%) alleles from individuals of gnomAD v4 SouthAsian background and has not been observed (not in any individuals) in 1kG All.

Clinical and Literature evidence: This variant has been classified as pathogenic in ClinVar database (Variation ID: [418295](#)). This variant has been previously reported in patients affected with autosomal recessive cone-rod dystrophy in both homozygous and compound heterozygous state (PubMed: [31006324](#)).

In-silico prediction: The *in-silico* predictions of the variant are damaging by SIFT, PolyPhen2 (HumDiv), LRT and deleterious by MutationTaster2. The reference codon is conserved across mammals in PhyloP and GERP++ tools.

OMIM phenotype: Ceroid lipofuscinosis, neuronal, 7 (OMIM#[610951](#)) is caused by homozygous or compound heterozygous mutation in the *MFSD8* gene (OMIM#[611124](#)). Macular dystrophy with central cone involvement (OMIM#[616170](#)) is caused by compound heterozygous mutation in the *MFSD8* gene (OMIM#[611124](#)). The neuronal ceroid lipofuscinoses (NCL, or CLN) are a clinically and genetically heterogeneous group of neurodegenerative disorders characterized by the intracellular accumulation of autofluorescent lipopigment storage material in different patterns ultrastructurally. These diseases follow autosomal recessive pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as likely pathogenic variant. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of the finding. If the results do not correlate, additional testing may be considered based on the phenotype observed.**

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re - alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes

Total Read Generated	14.66 Gb
Data ≥ Q30	96.53%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*,

GBA, and *CYP21A2* which have very high homology. Moreover, the *ORF15* transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.

- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

Reference

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in medicine: official journal of the American College of Medical Genetics*. 17.5 (2015): 405 -424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). *Nucleic Acids Res*. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." *Genetics in medicine: official journal of the American College of Medical Genetics* vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

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